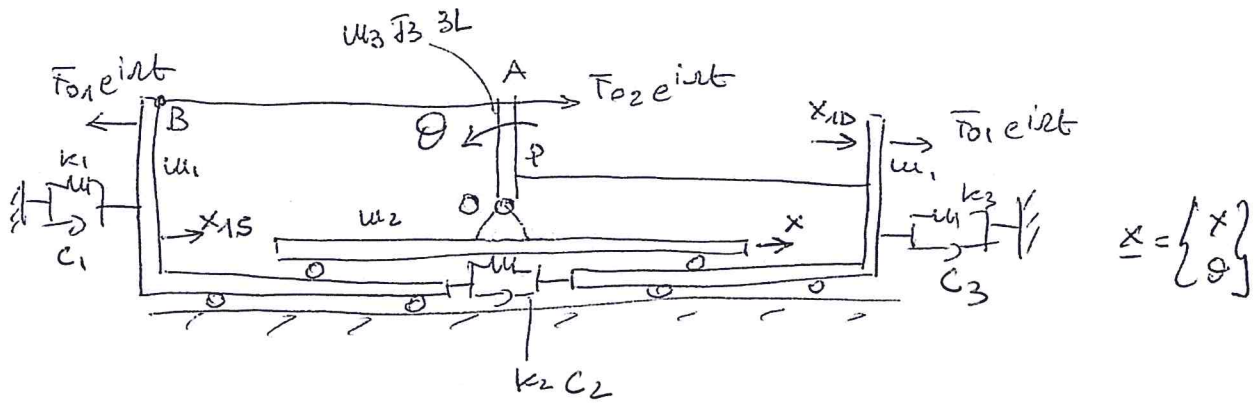


Written Test 17th June 2019 Simplified problem



4 rigid bodies $\rightarrow$ 12 d.o.f.

6 constraints + 1 hinge + 2 strings $\rightarrow$ 10 d.o.c.

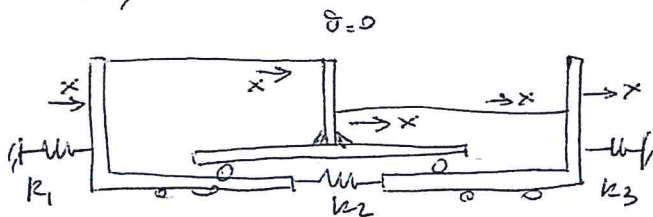
$\rightarrow$ 2 d.o.f.

$$T = \frac{1}{2} m_1 \dot{x}_{15}^2 + \frac{1}{2} m_2 \dot{x}^2 + \frac{1}{2} m_1 \dot{x}_{10}^2 + \frac{1}{2} m_3 \dot{x}_4^2 + \frac{1}{2} I_3 \dot{\theta}^2$$

$$D = \frac{1}{2} \sum_{i=1}^3 C_i \Delta L_i^2 ; \quad V_{\text{el}} = \frac{1}{2} \sum_{i=1}^3 k_i \Delta L_i^2 ; \quad V_g = m_3 g h_{g3}$$

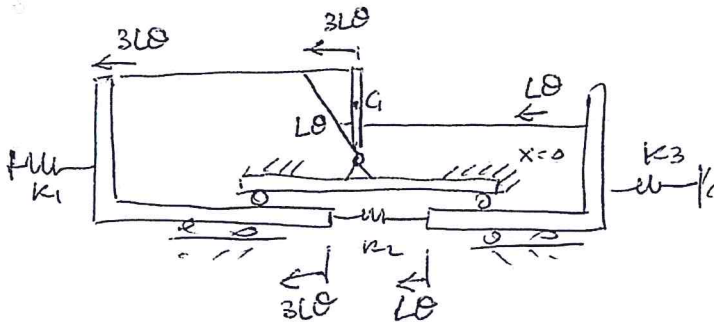
$$\delta W = F_{01eint} \delta x_{10} - F_{01eint} \delta x_{15} + F_{02eint} \delta x_A$$

$x \neq 0 ; \theta = 0$



$$\begin{aligned} x_{15} &= x & \Delta L_1 &= x \\ x_{10} &= x & \Delta L_2 &= 0 \\ x_4 &= x & \Delta L_3 &= -x \\ x_A &= x \end{aligned}$$

$x = 0 ; \theta \neq 0$



$$\begin{aligned} x_{15} &= -3L\theta & \Delta L_1 &= -3L\theta \\ x_{10} &= -L\theta & \Delta L_2 &= 3L\theta - L\theta = 2L\theta \\ x_4 &= -3/2\theta & \Delta L_3 &= L\theta \\ x_A &= -3L\theta \end{aligned}$$

$$w_{PH} = \text{diag}(w_1, w_2, w_1, w_3, \theta_3)$$

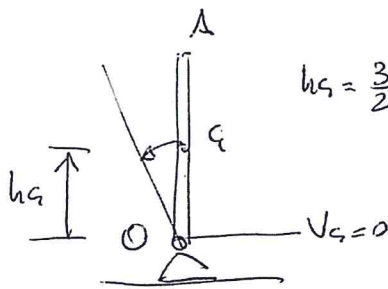
$$c_{PH} = \text{diag}(c_1, c_2, c_3)$$

$$k_{PH} = \text{diag}(k_1, k_2, k_3)$$

$$\bar{F} = \{ \bar{F}_{01} \quad -\bar{F}_{01} \quad \bar{F}_{02} \}^T$$

$$\Delta w = \begin{bmatrix} 1 & -3L \\ 1 & 0 \\ 1 & -L \\ 1 & -\frac{3L}{2} \\ 0 & 1 \end{bmatrix}$$

$$\Delta k = \Delta c = \begin{bmatrix} 1 & -3L \\ 0 & 2L \\ -1 & L \end{bmatrix}$$



$$h_g = \frac{3}{2} L \cos \theta \rightarrow k_g = w_3 g \begin{bmatrix} 0 & 0 \\ 0 & \frac{3}{2} L \end{bmatrix}$$

$$\Delta d = \begin{bmatrix} 1 & -L \\ 1 & -3L \\ 1 & -3L \end{bmatrix}$$

$$\sum F_x = 0 :$$

$$T_{AB} = \bar{F}_{el1} - \bar{F}_{el2} + w_1 \ddot{x}_{15}$$

